X-ray diffraction imaging system for the detection of illicit substances using pixelated CZT-detectors

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***Abstract*–The use of energy-dispersive X-ray diffraction (XRD) in security screening applications allows for highest detection and lowest false-alarm rates. So far, only Germanium-detectors have been used in commercially available XRD baggage-scanning systems, limiting the available detection area due to technical and commercial reasons, and as a consequence, the achievable throughput. Morpho Detection has built an innovative X-ray diffraction imaging (XD*i*) system equipped with 17 pixelated Cadmium-Zinc-Telluride (CZT) detector modules for increased detection-area to alleviate this limitation. A multi-focus X-ray source (MFXS) allows for high X-ray flux and makes any other mechanical movement, except for the bags, obsolete. In conjunction with a dedicated collimation system the detectors measure the scatter from the bag spatially and energetically resolved, allowing for a voxelized reconstruction of the bag's scatter-signature. The detector-array is required to deliver spectra of high quality, calling for very good energy-resolution and efficiencies, which is why signal-corrections are performed on-board. Each module is a completely independent unit to ensure service friendliness, receiving just one voltage, a trigger and Ethernet for control and data-transfer. We will explain the functional principle of the XD*i* -system, present the performance of the detectors with respect to stability and show spectra of different materials and reconstructed 3-D feature maps.**

I. Introduction

X-RAY diffraction (XRD) is a powerful method for screening airport baggage with respect to containing illicit substances such as commercial, military and home-made explosives, which are mostly polycrystalline and therefore give very distinct diffraction signals [1], but also liquids, aerosols and gels have an internal molecular order, and therefore produce characteristic diffraction signals as well [2].

XRD probes the structure of a material by measuring the coherently scattered rays that obey Bragg's law, which states that constructive interference can be observed from diffracted waves of wavelength λ under the scattering angle Θ, if the phase-difference of the waves diffracted at lattice planes with lattice constant d is a multiple n of the wavelength λ [3]:

$$n\lambda = 2d \sin\frac{\Theta}{2} \quad (1)$$

Substituting λ by the relation $E=hc/\lambda$, with h being Planck's constant and c the speed of light, and rewriting (1) yields (2),

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Fig. 1: Picture of the XD*i* cabin-baggage screener.

$$\frac{E}{hc}\cdot\sin\frac{\Theta}{2} = \frac{n}{2d} \quad (2)$$

which infers, that for constant energy E, signals from different lattice spacings d can be measured at different scattering angles (monochromatic method), or, the same can be measured by using a broad band of incoming energies and measuring the energy-resolved signal under constant angle Θ.

This so-called Laue- or energy-dispersive method is employed in baggage-inspection, since only sources with a broad bremsstrahlungs-spectrum, like X-ray tubes, provide enough photon-flux to inspect bags in a reasonable amount of time.

Due to XRD's material specificity, it allows for high probability of detection at low false alarm rates. So far, only two commercial products have existed on the market, the first one introduced in 1999 by Heimann using a pencil-beam geometry [4]. Its obvious drawback was the limited object-coverage, which is why it was used for clearing false-alarms at specific locations in a bag only. The second system was introduced in 2000 by Yxlon as XES3000 employing a cone-beam scatter-geometry for increased object-coverage allowing for complete bag-scans in about 60s. Its successor, the XRD3500 [5], employs the same diffraction geometry, but X-ray source and detector are not rigidly coupled via a big C-arm, but instead are electronically geared via linear motors. It has an imaging stage for identification of threat regions detected by other up-stream scanners, and, hence, is mostly used for clearing false-alarm bags as well. Despite certification by the most demanding authorities for highest detection and lowest false-alarm rates, these systems have never been sold in high volume because of their high price at given throughput. Cost factors are the precise and bulky drive-mechanics

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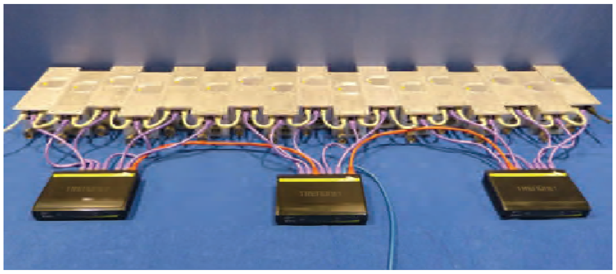

Fig. 2: Array of 17 CZT-modules with trigger and ethernet cables connected.

associated with the need for a big and heavy led-plated cabin, precision collimators, a 12kW rotating-anode tube, and, as the most expensive single part, a 100 mm diameter Germanium-detector.

In order to address the problems of cost and throughput, a completely new geometry has been proposed that just requires the movement of the bag, eliminating the bulky mechanics required for alignment of source and detector. The concept foresees stationary components only, i.e. a multi-focus X-ray source (MFXS) allowing for high instantaneous focal spot power with efficient anode cooling, and an array of room-temperature semi-conductor detectors (cf. Fig. 2) to maximize detection area and thus the capture of scatter photons to increase throughput [6]. This innovative concept allows to build an X-ray diffraction imaging (XD*i*) system that is so significantly reduced in size as compared to the XRD3500, that it can operate as cabin-baggage scanner at a security checkpoint (cf. Table 1).

TABLE I. COMPARISON OF XRD3500 AND XD*i*

Property	Unit	XRD3500	XD*i*
Length	m	7	3.3
Width	m	2.5	1.65
Height	m	2.5	2
Weight	t	9.8	2
Tunnel	cm²	100 x 52	62 x 42
Max. Power	kW	25	10

II. FUNCTIONAL CONCEPT OF XDI

The MFXS is designed to run at a power of 6 kW and has 16 focal spots that fire sequentially for 125 µs, repeating the cycle every 2.1 ms. A dedicated primary collimation system lets 17 small beamlets pass each time a focus fires, so that after one complete focus-sequence a web of X-rays has swept through a 0.21 mm thick slice of a bag when that one is moving at a speed of 10 cm/s. The beamlets nearly ensure full coverage of the whole tunnel section (cf. Fig 3, left). Some X-rays interact with the object generating scatter-radiation, which eventually passes through the slits of the secondary collimator under the scattering-angle defined by system-geometry. The secondary collimator principally consists of a stack of thin metal plates of 1.3 m length underneath the conveyor and images the scatter from the object topographically onto the 17 detector-modules' pixel-lines (cf. Fig. 3, right). In this way the object's diffraction signature is measured spatially and energetically resolved. Whenever a focus fires, a trigger-logic sends focus-number and belt-position to all detectors via a daisy-chain, initiating a data-acquisition period that ends after 125 µs. All detected photons are immediately corrected for charge-sharing, depth of interaction, energy-gain and offset, are accumulated in a data-package together with trigger-information, module-number, pixel-location and photon energy-value, to then be sent to the host-PC via Ethernet. The photon-information arriving from the detector-array is pre-processed, reconstructed into the voxels that the photons are originating from, and then post-processed for threat identification.

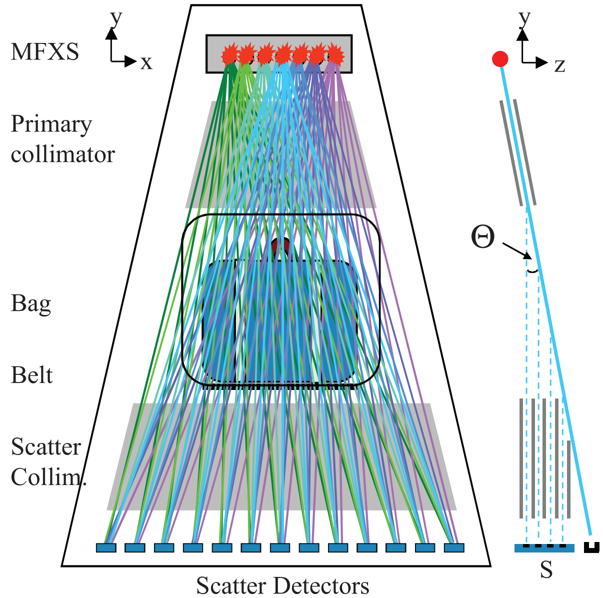

Fig. 3: Left: Schematic of XD*i* set-up, indicating only 7 of the 16 foci and 13 of the 17 detector-modules. The bag is transported into the z-direction. Also shown are the meshs of beamlets that would sequentially penetrate the object space when the foci are switched on (here all at once). Right: View onto the diffraction plane. At different heights of object-space scatter is generated and seen by those lines of detector S that look through the secondary collimator into the respective direction.

III. CZT- INSTEAD OF GERMANIUM-DETECTORS

The most compelling reason for employing room-temperature semi-conductors is the need for having detector-areas distributed over a width of 130 cm. This would be costly to realize with Ge-detectors, not only for the price of detectors and associated read-out electronics alone, but also because of the required large vacuum-vessel, or several smaller ones, and sufficient, reliable, cooling power for keeping cryogenic temperatures. An easy and quick exchange of such a detector seems impossible, and therefore is not suitable for a security checkpoint environment. This is also the reason, why our CZT-detector array consists of completely autonomous sub-units for imaging diffraction signatures. They just need a 12V supply-voltage and a trigger input for synchronization with the source. Both are daisy-chained among the modules. Control and data-transmission are realized via Ethernet. The modules have been developed by Leti-Minatec for Morpho Detection and consist of a pixelated CZT-crystal, although spatial resolution is only required in z-direction, but of course the pixelation is necessary for utilizing the small pixel-effect [7]. Anodes are read out by the IdeF-X –ASIC [8] and cathodes by the ALIGASPECT-ASIC (designed by Leti). Heart of the system is an FPGA running a NIOS operating system, controlling all read-out and performing on the fly charge sharing and depth of interaction (DOI) corrections [9]. Characteristics of the module are given in table 2 or can be found in [10]. In order to maintain stable performance, the modules are held at a temperature of 22±1°C.

TABLE II. CHARACTERISTICS OF IMADIF MODULE

Key Feature	Characteristic
Crystal	CZT 5mm
Number of pixels	192
Number of lines	22
Anode read-out ASIC	IDeF-X
Cathode read-out ASIC	ALIGASPECT
Avg. energy-resolution at 59.5 keV (40 modules)	2.26 keV± 0.13 keV (σ)
Peak-efficiency (59.5±3keV)	73±4% (σ)

IV. CALIBRATION OF MODULES

The detector-module has on-board calibration capability for energy- and DOI-calibration, which is performed in the last manufacturing stage with Am^{241}- and Co^{57}-sources.
However, once the modules are installed in the XD*i*-system, sources cannot be used efficiently due to the tight collimation in front of them, and sources would probably not be tolerated in an airport environment. Hence, for calibration in the field the tungsten characteristic K_α-line at 59.31 keV is used, dominant in the X-ray tube spectrum.

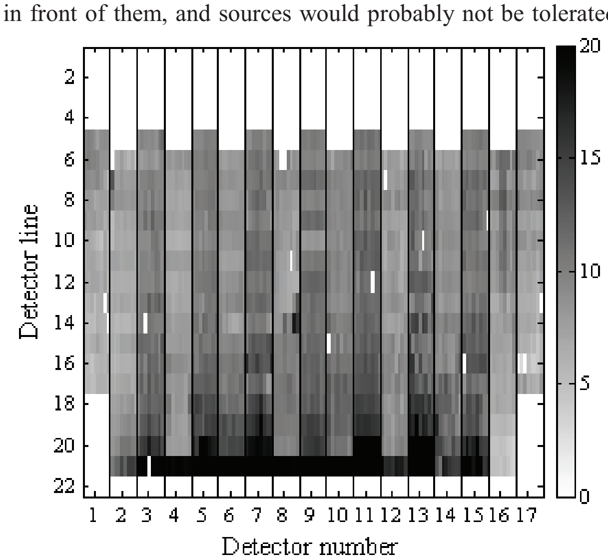

Fig. 4: Intensity on the array of 17 detectors in counts/mm²/s on each respective pixel integrated over all foci and 10-140keV. High intensities in detector line 21 originate from the conveyor-belt. Pixels that receive scatter from outside the tunnel-region are not shown and are white, as well as dead pixels. Inbetween is the scatter from the foam-cube.

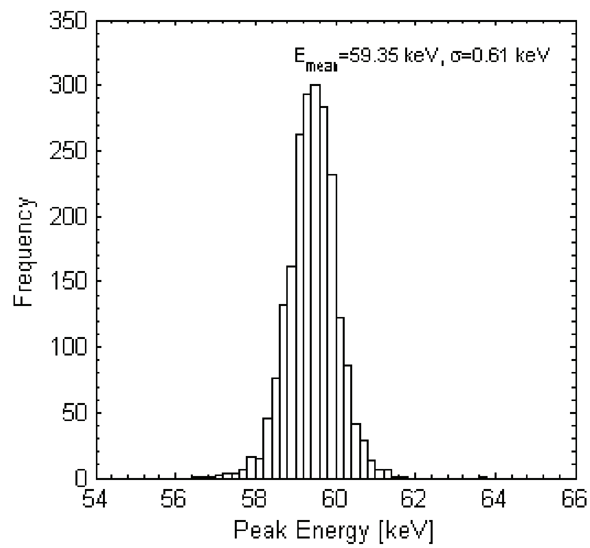

Fig. 5: Histogram of peak-positions on 2175 pixels considering all values between 55 to 65 keV. 89 pixels are out of this range.

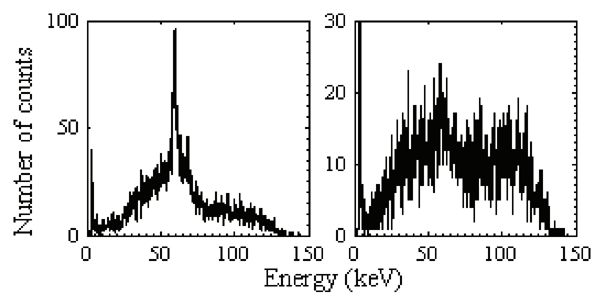

Fig. 6: Example of two pixel spectra with high and low number of counts.

We irradiate a foam-cube to simultaneously generate a scatter-signal in each height of object space, and consequently on each relevant detector line that looks into object space. The resulting intensity in cts/mm²/s for each pixel, integrated over all foci, energy-range from 10 to 140keV for a 600s measurement is shown in Fig. 4. Pixels that don't look into object-space or are defective have been drawn white. High scatter-intensities in detector lines 20 and 21 originate from the conveyor-belt. Lines from 5 to 20 show the scatter-intensities of the foam-cube with an average of 10 cts/mm²/s. Differences in intensity between pixels can be explained by the fact that not all regions in object space are equally often hit by a beamlet during the sweep, as well as variations in CZT-detector performance.
In energy-calibration mode the detector-module uses such a measurement to perform its respective calibration. The algorithms are not perfect, yet, mainly suffering from low statistics spectra at the edges of object-space that make peak-determination prone to errors. Two pixel-spectra that the algorithm operates on are shown in Fig. 6. A histogram of

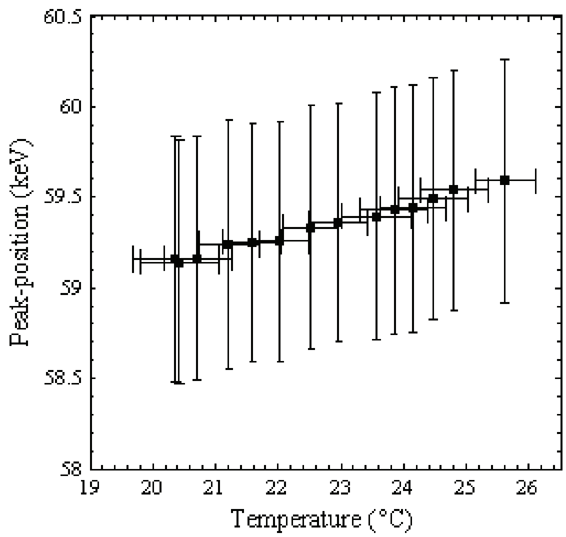

Fig. 7: Detector-array peak-positions at different module temperatures. For each measurement the peak-positions determined on 2175 pixels are averaged and plotted against the average of the 17 modules internal temperature. The bars indicate the standard deviation. The resulting temperature-gradient for the peak-shift corresponds to 0.1 keV/°C.

those pixels that look through the secondary collimator into object-space (2264) is shown in Fig. 5. Mean and standard deviation have been calculated from pixels with a peak-position between 55 and 65 keV. 89 pixels are out of this range due to wrongly determined peak-position, assumably caused by low statistic spectra, or simply because they are dead. The mean value is with 59.35 keV close to the target of 59.31 keV at a standard deviation of 0.61 keV.

V. Temperature Stabilty of calibration

The modules have been developed for operating at temperatures from 20-25°C with a target at 22°C. A long cooling-plate attached to the modules from underneath is keeping them at this temperature. However, in order to investigate the modules' peak-stability within the abovementioned temperature-range, the cooling-plate temperature is varied in about 0.5°C steps to generate different internal module-temperatures that can be measured with a dedicated sensor. At each temperature a foam-cube measurement is conducted and peak-positions are determined from spectra as shown in Fig. 6. The temperature-values and peak-positions are averaged for all modules at each cooling-plate temperature. The resulting measurements are shown in Fig. 7 together with their standard deviations. A peak-shift gradient of 0.1 keV/°C can be deducted, which means that module temperature variations within the specified range are acceptable.

Long-term stability of detector-characteristics is currently under investigation and is important in determining the necessary interval for repeating calibration or replacing modules.

Fig. 8: Bag prepared for scanning with 4 substances, acetone, water, acrylic glass (PMMA) and salt.

VI. First bag-scan

The XD*i*–system is a fully functional prototype unit allowing to routinely scan bags. A bag is prepared with four different substances, a water-bottle and acetone-bottle above a pack of salt and a cuboid of acrylic glass (PMMA) (Fig. 8), and then screened with the XD*i* –system. Knowing focus-number, belt-position and pixel-position for each photon, the topographically measured scatter-data can be back-projected into the voxelized object-space, reconstructing its scatter-signature in all three dimensions in this way. A 3D-representation of the bag's scatter-signal strength is rendered from the voxels and shows the four objects and faintly the wheels and walls of the bag (cf. Fig 9).

Adding up the voxels' spectra over the regions of interest and available slices yields the characteristic momentum-spectra of the four materials that can already be distinguished by eye (cf. Fig. 10). Automatic threat-detection software will operate on these spectra to identify threats.

Fig. 9: Three-dimensionally reconstructed distribution of scatter-strength of the scanned bag. . Top left is water, right acetone, bottom left is PMMA and right salt. The wheels and walls of the bag can also be identified.

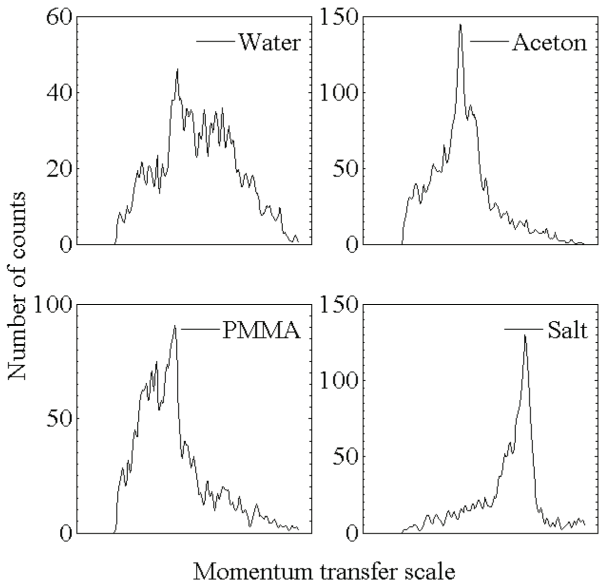

Fig. 10: Several voxels of each object in the bag are summed up to yield the spectra as shown above.

VII. CONCLUSION

A novel baggage-screening system for carry-on baggage based on X-ray diffraction has been developed. Its photon yield at 0.6kW tube-power per 1 cm/s belt-speed is sufficient to scan bags and to reconstruct the bag's contents' scatter-signatures for threat-identification.

The increase in photon-flux as compared to the XRD3500 is achieved by a more efficient diffraction geometry, a high-power MFXS and an increase in detector-area by a factor 4.5 by employing 17 CZT-detector modules, of which each works as an autonomous unit. The modules' average peak-stability with temperature is 0.1 keV/°C within the range of 20-25°, and thus gives no reason to worry about temperature drifts in that range.

We have demonstrated in a bag-scan that photon-flux and spectral quality are sufficient to generate data that allows for distinction of liquids and crystalline substances. Also, we have used the spatially resolved scatter-information to reconstruct the 3-dimensional scatter-signature of the bag's objects.

We are currently working on industrializing the XD*i*-system and realizing the automation of threat detection.

VIII. ACKNOWLEDGEMENTS

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